

Infinite paths in the composite-restricted visible lattice

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Abstract

We consider the nearest-neighbour graph on pairs of integers greater than one that are coprime and have at least one composite coordinate. We prove that this graph contains an infinite simple path, answering Erdős Problem 1212 affirmatively. More precisely, for almost every slope in a fixed interval away from the diagonal, there is a ray with that limiting coordinate ratio. The proof first obtains many pairwise coprime composite rows free of small prime factors in a short band. A polynomial Jacobsthal bound supplies composite supporting columns, so a bad crossing through the band has a subcrossing of polylogarithmic width. Its large common prime divisors then force an integer interpolation polynomial of small height to vanish at one of its vertices. A summable estimate for the exceptional directions of all such polynomials permits one direction to be fixed at every sufficiently large scale. Planar crossing duality and explicit overlapping rectangles produce an unbounded connected subgraph along this direction, from which an infinite simple path is extracted.

1 The graph and the main theorem

Let

$$\mathcal{S} = \{(x, y) \in \mathbb{N}^2 : x, y > 1, \gcd(x, y) = 1, \text{ at least one of } x, y \text{ is composite}\}. \quad (1)$$

Two vertices are adjacent when one coordinate changes by one and the other does not change. A ray is a sequence of distinct vertices with consecutive vertices adjacent. Every ray in this locally finite graph leaves every finite set.

Erdős asked whether this graph has a path going to infinity [1, p. 114]. The composite-coordinate condition is important: Stewart's construction for the unrestricted visible lattice uses vertices whose two coordinates are prime. Vardi studied the infinite component of the unrestricted visible lattice and its asymptotic density [5]. Random-coset versions of coprime percolation have also been studied [3]. Here every path is constructed in the deterministic graph (1).

Theorem 1.1. *For Lebesgue-almost every $\alpha \in (4/3, 5/3)$ there is a ray $((x_t, y_t))_{t \geq 0}$ in $\mathcal{S}$ such that*

$$\frac{x_t}{y_t} \longrightarrow \alpha.$$

In particular, the graph on $\mathcal{S}$ has an infinite path.

For example, $(4, 9), (5, 9), (5, 8)$ is a finite path in $\mathcal{S}$. The vertex $(4, 7)$, however, is isolated: its horizontal neighbours have two prime coordinates, and its vertical neighbours are nonvisible. Thus the theorem asserts an infinite component in a graph that also contains isolated vertices.

The central step is to associate a small integer polynomial to a long *bad* crossing, where a point with both coordinates greater than one is bad when it is outside $\mathcal{S}$. An integer is called *z-rough* when all its prime factors exceed z . We obtain a short band containing many

rough composite rows. Safe vertical supports localize a bad subcrossing in that band. At every selected row, a bad vertex has a large common prime divisor; the divisors are distinct. An interpolation polynomial in the short offsets then has an evaluation divisible by their product. This product exceeds the size of a nonzero evaluation, so the evaluation is zero. The polynomial may depend on the crossing. We handle this dependence by excluding the directions associated with *all* polynomials of the required degree and height at once. Finally, overlapping rectangles in a single narrowing corridor connect the crossings at successive scales.

2 Notation and preliminary facts

An integer interval $[a, b]_{\mathbb{Z}}$ contains both endpoints and has $b - a + 1$ sites. An integer rectangle is a product of two such intervals. A left-right crossing is a path inside the rectangle joining its left and right sides; top-bottom crossings are defined similarly. Safe paths use the four nearest neighbours. A bad star path may also use a diagonal step, so the maximum of the two coordinate changes is one. We write $\omega(n)$ for the number of distinct prime divisors of n , $\mu(n)$ for the Moebius function, and p_k for the k th prime, with $p_1 = 2$.

Lemma 2.1 (Finite crossing facts). *In an integer rectangle, a safe left-right crossing exists if and only if there is no bad top-bottom star crossing. Furthermore, any nearest-neighbour left-right and top-bottom crossings of the same rectangle have a common vertex.*

Proof. These are the finite four/eight-neighbour crossing alternatives. One way to obtain the alternative is to join an auxiliary left side to every safe site reachable from the left, and trace the boundary on the right of this reachable set. If the set does not reach the right side, the boundary has a segment joining the upper and lower sides. The lattice sites just outside it are bad; successive such sites share an edge or a corner and give the required star crossing. Repetitions can be removed. Conversely, a left-right lattice path separates the upper and lower sides after its ends are extended outside the rectangle. A top-bottom star crossing must meet it. A diagonal edge between lattice sites cannot cross a horizontal or vertical lattice edge in its interior without a common lattice endpoint, so such an intersection would be a site of both colours. This is impossible. The same separation argument for two nearest-neighbour crossings proves the last assertion. $\square$

We use the standard estimates

$$\prod_{p \leq z} (1 - p^{-1}) \sim \frac{e^{-\gamma}}{\log z}, \quad \sum_{z < p \leq 2z} \frac{1}{p} \sim \frac{\log 2}{\log z}, \quad \sum_{p \leq z} \frac{1}{p} = \log \log z + O(1), \quad (2)$$

and $p_k = O(k \log(k + 1))$. These follow from the prime number theorem, Mertens' theorem and partial summation; see [4]. We use Vaughan's uniform Jacobsthal bound [2, p. 329]: for any set Q of k primes, every interval of

$$O((k + 1)^2 \log^4(k + 2)) \quad (3)$$

consecutive integers contains an integer coprime to $\prod_{q \in Q} q$. The constant is absolute and the primes in Q are arbitrary.

Throughout the proof N is sufficiently large, and we write

$$\ell = \log N, \quad L = \log \ell, \quad u = 1000L, \quad z = N^{1/u}, \quad d = \lceil 4u \rceil, \quad r = \binom{d+2}{2} - 1. \quad (4)$$

The numerical parameter comparisons and counting estimates hold above a single absolute lower bound for N , uniformly in interval positions, chosen rows and bad paths. Later, the scale after which a fixed direction avoids the exceptional sets is allowed to depend on that direction.

3 Rough composite rows in short bands

Lemma 3.1. *Every integer interval $J \subset [N/2, 4N]$ with $T = |J| \geq N^{9/10}$ contains at least*

$$\frac{T}{100(\log z)^2} \quad (5)$$

composite integers whose prime factors all exceed z .

Proof. Put $P = \prod_{p \leq z} p$, $V = \prod_{p \leq z} (1 - 1/p)$, and $\mathcal{Q} = \{q \text{ prime} : z < q \leq 2z\}$, with $S = \sum_{q \in \mathcal{Q}} q^{-1}$. For an integer h coprime to P , let

$$A_h = \#\{t \in J : h \mid t, \gcd(t, P) = 1\}.$$

Choose an odd integer $m = 100L + O(1)$. Bonferroni's inequalities at orders m and $m+1$ bound A_h below and above by

$$\sum_{\substack{a|P \\ \omega(a) \leq m}} \mu(a) \#\{t \in J : ha \mid t\}, \quad \sum_{\substack{a|P \\ \omega(a) \leq m+1}} \mu(a) \#\{t \in J : ha \mid t\},$$

respectively. Each count is $T/(ha) + O(1)$, with absolute error at most one. The gap between the corresponding truncated main coefficients and V is at most the elementary symmetric sum of order $m+1$ in the numbers $1/p$, $p \leq z$. By (2), their sum is at most $2L$ eventually. Thus the coefficient error is at most

$$\varepsilon_N = \left(\frac{2eL}{m+1} \right)^{m+1} \leq \ell^{-200}. \quad (6)$$

The number of divisors retained at either order is at most $E_N = (m+2)z^{m+1}$. We have obtained, uniformly in h coprime to P ,

$$\frac{T}{h}(V - \varepsilon_N) - E_N \leq A_h \leq \frac{T}{h}(V + \varepsilon_N) + E_N. \quad (7)$$

Apply the first two inclusion-exclusion terms to divisibility by primes in $\mathcal{Q}$. The number of integers counted by at least one of these events is at least

$$\sum_{q \in \mathcal{Q}} A_q - \sum_{\substack{q, q' \in \mathcal{Q} \\ q < q'}} A_{qq'}.$$

Using (7) and $\sum_{q < q'} (qq')^{-1} \leq S^2/2$, this is at least

$$TV(S - S^2/2) - T\varepsilon_N(S + S^2/2) - O(N^{1/5}). \quad (8)$$

Indeed, the summed integer-count remainder is at most $4(m+2)z^{m+3} \leq N^{1/5}$ eventually, because $(m+3)/u \rightarrow 1/10$.

By (2), the main coefficient in (8) is asymptotic to $e^{-\gamma} \log 2 / (\log z)^2$. Both errors are $o(T/(\log z)^2)$ uniformly for $T \geq N^{9/10}$. This proves (5). Finally, every counted integer exceeds $N/2 > 2z$ and has a divisor $q \in \mathcal{Q}$, so is composite. $\square$

Lemma 3.2. *Set $B = \lceil 10^4 \ell^2 \rceil$. Every interval J as in Lemma 3.1 contains a block J_0 of exactly B consecutive integers and distinct composite integers $c_1, \dots, c_r \in J_0$ whose prime factors exceed z . The c_i are pairwise coprime.*

Proof. Partition J into full blocks of B integers and a final shorter block. Removing the latter loses at most $B = o(T/(\log z)^2)$ selected integers. By averaging, one full block contains at least

$$\frac{B}{200(\log z)^2} \geq 50u^2$$

of them. Since $r = 8u^2 + O(u)$, this is sufficient. Also $B < z$ eventually. A common prime factor of two selected integers would exceed z but divide a nonzero difference smaller than B , an impossibility. $\square$

4 Localizing a bad crossing

Lemma 4.1. *Let $J_0 \subset [N/2, 4N]$ be a block of B consecutive integers and put $D_N = \lceil \ell^{11} \rceil$. On either side of each $x \in [N/2, 4N]$, within distance D_N , there is a composite integer v such that $\{v\} \times J_0 \subset \mathcal{S}$.*

Proof. Let Q_0 be the set of all prime divisors of the integers in J_0 . Its cardinality k satisfies

$$k \leq B \frac{\log(4N)}{\log 2} = O(\ell^3).$$

Choose a prime $q \notin Q_0$ with $q \leq p_{k+1}$. By (3), integers a coprime to $\prod_{p \in Q_0} p$ occur with gaps $O((k+1)^2 \log^4(k+2))$. Consequently integers $v = qa$ of this kind occur with gaps $O((k+1)^3 \log^5(k+2))$. For v near the indicated x we have $a > 1$, because $q = N^{o(1)}$ and $x \geq N/2$. Hence v is composite. It is coprime to every element of J_0 , giving the required safe segment. The gap bound is $O(\ell^9(\log \ell)^5) = o(\ell^{11})$; allowing an additional endpoint gap yields a support strictly on either side within D_N for all sufficiently large N . $\square$

Lemma 4.2. *A bad top-bottom star crossing of a rectangle $I \times J \subset [N/2, 4N]^2$ with $|J| \geq N^{9/10}$ contains points*

$$v_i = (x_0 + a_i, y_0 + b_i), \quad 1 \leq i \leq r,$$

where (x_0, y_0) is a point of that crossing, $|a_i|, |b_i| \leq R_N := 2D_N + B$, and there are distinct primes $q_i > z$ dividing both coordinates of v_i .

Proof. Choose J_0 and c_i by Lemma 3.2. Orient the crossing from bottom to top. Take its first visit to the upper row of J_0 and its last preceding visit to the lower row. The intervening subpath stays inside the closed band: a visit above its upper row would contradict the first choice, and a visit below its lower row would force a later return to the lower row. Let (x_0, y_0) be its first vertex.

The two safe support columns from Lemma 4.1 cannot be crossed by this bad subpath. A star step changes the first coordinate by at most one, and the subpath cannot pass around an endpoint while remaining in the band. Its horizontal span is therefore at most $2D_N$. It visits every selected row c_i ; choose v_i on that row. Since c_i is composite, badness at v_i implies nonvisibility, and a prime q_i dividing the common divisor exceeds z . The q_i are distinct because the c_i are pairwise coprime. The displayed offset bounds follow from the horizontal span and band height. $\square$

5 The interpolation equation

The height of an integer polynomial is the largest absolute value of its coefficients. The following elementary interpolation step keeps track of height as well as degree.

Lemma 5.1. *Let $d \geq 1$, $r = \binom{d+2}{2} - 1$, and $R \geq 1$. For any r integer pairs (a_i, b_i) with $|a_i|, |b_i| \leq R$ there is a nonzero $H \in \mathbb{Z}[A, B]$ of total degree at most d such that*

$$H(a_i, b_i) = 0 \quad (1 \leq i \leq r), \quad \text{ht}(H) \leq r!R^{dr}. \quad (9)$$

If distinct primes q_i divide both coordinates of $(x_0 + a_i, y_0 + b_i)$, then

$$\prod_{i=1}^r q_i \mid H(-x_0, -y_0). \quad (10)$$

Proof. Evaluate the $r + 1$ monomials of total degree at most d at the given points. This gives an r by $r + 1$ integer matrix whose entries have absolute value at most R^d . Let its rank be $t \leq r$. Choose t independent rows, t columns forming a nonsingular minor, and one further column. The signed minors of the resulting t by $t + 1$ matrix give a nonzero integer kernel vector. Extend it by zero on all other columns. It annihilates the chosen rows, hence every row. Each coefficient is at most $t!R^{dt} \leq r!R^{dr}$. This proves (9), without a full-rank assumption.

Modulo q_i the offset point equals $(-x_0, -y_0)$. Hence q_i divides $H(-x_0, -y_0)$. Distinctness gives (10). $\square$

Proposition 5.2. *Set $\mathcal{H}_N = \lceil \exp((\log \log N)^6) \rceil$. Every bad top-bottom star crossing of a rectangle $I \times J \subset [N/2, 4N]^2$ with $|J| \geq N^{9/10}$ contains a point (x_0, y_0) satisfying $F(x_0, y_0) = 0$ for some nonzero $F \in \mathbb{Z}[X, Y]$ of total degree at most d and height at most $\mathcal{H}_N$. The same statement holds for a left-right bad star crossing when $|I| \geq N^{9/10}$.*

Proof. Apply Lemmas 4.2 and 5.1. Because $R_N = O(\ell^{11})$ and $d = O(L)$, $r = O(L^2)$, we have

$$\log(r!R_N^{dr}) = O(L^4) < L^6 \quad (11)$$

eventually. Also

$$\log \prod_i q_i > \frac{r}{u} \ell, \quad |H(-x_0, -y_0)| \leq (r + 1) r! R_N^{dr} (4N)^d.$$

The logarithm of the latter bound is $d\ell + O(L^4)$, whereas $r/u = 8u + O(1)$ and $d = 4u + O(1)$. Thus the prime product strictly exceeds the bound for a nonzero evaluation. The divisibility in (10) forces $H(-x_0, -y_0) = 0$. Set $F(X, Y) = H(-X, -Y)$.

For a horizontal bad crossing, interchange the coordinate roles in the band and support construction. The interpolation applies to the original coordinate pair, or equivalently its resulting polynomial is transposed back. The point still lies on the original crossing and the degree and height bounds do not change. $\square$

6 A fixed direction avoiding the equations

For dyadic $N = 2^j$, define

$$\rho_N = \exp \left(-\frac{\log N}{20000 \log \log N} \right). \quad (12)$$

Consider all nonzero polynomials $f \in \mathbb{Z}[T]$ of degree at most d and height at most $\mathcal{H}_N$. Around the real part of each complex root put an interval of radius $4\rho_N$, and let E_N be their union. Constant polynomials contribute no intervals. Writing $|E_N|$ for Lebesgue measure, we have

$$|E_N| \leq 8\rho_N d(2\mathcal{H}_N + 1)^{d+1}. \quad (13)$$

Its logarithm is bounded above by

$$-\frac{\log N}{20000 \log \log N} + O((\log \log N)^7).$$

Therefore $\sum_j |E_{2^j}| < \infty$. For example, its summands are eventually bounded by j^{-2} . The measure of the set lying in infinitely many E_{2^j} is at most every tail of this convergent series, hence is zero.

Fix $\alpha \in (4/3, 5/3)$ outside that null set. It lies outside E_N for all sufficiently large dyadic N . Define the lattice corridor

$$\mathcal{U}_N(\alpha) = \{(x, y) \in \mathbb{Z}^2 \cap [N/2, 4N]^2 : |x/y - \alpha| \leq \rho_N\}. \quad (14)$$

Lemma 6.1. *No point of $\mathcal{U}_N(\alpha)$ is a zero of a nonzero integer polynomial of total degree at most d and height at most $\mathcal{H}_N$, for all sufficiently large dyadic N .*

Proof. Suppose $F(x, y) = 0$ at such a point. A nonzero constant cannot vanish, so let $m \geq 1$ be its total degree and put $f(T) = \sum_{a+b=m} F_{ab} T^a$, where F_{ab} is the coefficient of $X^a Y^b$. This is a nonzero integer polynomial of degree at most d and height at most $\mathcal{H}_N$. Since $1 < x/y < 2$ eventually and $y \geq N/2$, the lower-degree terms give

$$|f(x/y)| \leq \frac{2(r+1)2^d \mathcal{H}_N}{N} \leq N^{-1/2}$$

for all sufficiently large N . The factor 2^d is harmless because $d = O(\log \log N)$. If f is constant, this contradicts its being a nonzero integer. Otherwise factor it over $\mathbb{C}$. Its leading coefficient has absolute value at least one, so one of its roots β satisfies

$$|x/y - \beta| \leq N^{-1/(2 \deg f)} \leq N^{-1/(2d)} \leq \rho_N^2.$$

The last inequality uses $d \leq 5000 \log \log N$ eventually. It follows that $|\alpha - \Re \beta| \leq \rho_N + \rho_N^2 < 4\rho_N$, contradicting $\alpha \notin E_N$. $\square$

Corollary 6.2. *Every integer rectangle $I \times J \subset \mathcal{U}_N(\alpha)$ has a safe left-right crossing if $|J| \geq N^{9/10}$, and a safe top-bottom crossing if $|I| \geq N^{9/10}$, for all sufficiently large dyadic N .*

Proof. Failure of the first crossing gives a bad top-bottom star crossing by Lemma 2.1. Proposition 5.2 contradicts Lemma 6.1. Use the horizontal versions for the second claim. $\square$

7 Joining the scales

Proof of Theorem 1.1. Fix a direction α as in the preceding section, and take all dyadic scales sufficiently large for the preceding conclusions. Put

$$h_j = \left\lfloor \frac{2^j \rho_{2^j}}{100} \right\rfloor, \quad s_j = \lfloor h_j/10 \rfloor.$$

We have $h_j/(2^j)^{9/10} \rightarrow \infty$. For t large, the derivative of $t/(20000 \log t)$ lies between zero and $1/2$. Equation (12) therefore implies

$$\sqrt{2} < \frac{2N \rho_{2N}}{N \rho_N} < 2$$

eventually. In particular the integers h_j are nondecreasing and $h_{j+1} \leq 3h_j$ for all sufficiently large j .

At scale $N = 2^j$ choose the integer heights $N, N + s_j, N + 2s_j, \dots$ up to $2N$, adding the endpoint $2N$ if necessary. At each selected height t let $Q(j, t)$ be the integer square with centre $(\lfloor \alpha t \rfloor, t)$ and half-side h_j . Every point of this square satisfies

$$|x - \alpha y| \leq 1 + (1 + \alpha)h_j, \quad y \geq N - h_j.$$

Thus the square lies in $\mathcal{U}_N(\alpha)$, with ratio error less than $\rho_N/10$, eventually. Its coordinates lie in $[N/2, 4N]^2$. Choose a safe horizontal crossing of each square by Corollary 6.2.

For two consecutive squares Q, Q' at the same scale, their centres differ by at most $h_j/10$ vertically and $h_j/4$ horizontally, including the rounding of αt . Form a bridge rectangle T by intersecting their horizontal intervals and taking the union of their vertical intervals. Its horizontal side length is at least $7h_j/4$ and its vertical side length is at most $21h_j/10$, apart from the added endpoint site. The same ratio estimate, with h_j replaced by $5h_j/4$, puts T inside $\mathcal{U}_N(\alpha)$. Its horizontal side has more than $N^{9/10}$ sites, so it has a safe vertical crossing.

This vertical crossing meets the chosen horizontal crossing of each of Q, Q' . Indeed, restricting it between the lower and upper sides of Q gives a vertical subcrossing in the rectangle with Q 's vertical interval and T 's horizontal interval. Restricting the horizontal crossing of Q to that horizontal interval gives the opposite crossing of the same rectangle. Lemma 2.1 gives an intersection. The argument for Q' is identical. All horizontal crossings at this scale are connected.

At height $2N$, the last square at scale j and the first at scale $j+1$ have the same centre and half-sides $h_j \leq h_{j+1}$. Use a bridge whose horizontal interval is that of the smaller square and whose vertical interval is that of the larger. It lies in the larger square, hence in $\mathcal{U}_{2N}(\alpha)$. Since $2h_j + 1 > (2N)^{9/10}$ eventually, it has a safe vertical crossing by Corollary 6.2. The preceding intersection argument joins the horizontal crossings at the two scales.

The union of all these chosen paths is one connected unbounded subgraph of $\mathcal{S}$. Fix a vertex on the first horizontal crossing. It reaches arbitrarily large first coordinates in this same union. The tree of finite simple paths from this fixed vertex is finitely branching and has nodes of arbitrarily large depth; Koenig's lemma gives a ray in the union. Each finite collection of the early squares and bridges has finitely many vertices. On every sufficiently late one the coordinate ratio is within ρ_{2^j} of α . Since a ray eventually leaves every finite set and $\rho_{2^j} \rightarrow 0$, its ratio tends to α . $\square$

Corollary 7.1. *The graph on $\mathcal{S}$ has an infinite component in each of the regions $x > y$ and $x < y$. Removing finitely many vertices leaves an infinite component in each region.*

Proof. The construction gives a ray in $x > y$; transpose its coordinates for the other region. No nearest-neighbour path can change the sign of $x - y$ without visiting the excluded diagonal. A ray has only finitely many vertices in a prescribed finite set, so an appropriate tail survives its removal. $\square$

AI-assisted research

During the preparation of this work, the author used an author-developed local implementation of the Proof Engine infrastructure (development version 2.0) for AI-assisted research. The project-level research framework is described in *Proof Engine 2.0* [6]. See also *Proof Engine Infrastructure* (1.0) [7] for the complementary proof-development framework. The implementation uses replaceable AI-agent and execution-harness components; the present work used OpenAI Codex as its AI-agent component. The author is responsible for the definitions, statements, proofs, citations, code, and exposition.

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